

CHAMBER OF CURIOSITIES

The Door as a Gateway to Knowledge

CINEMATIC WebGL PORTFOLIO · COMPLETE DESIGN BRIEF

A mind-blowing immersive 3D experience that will leave every visitor speechless.

VERSION 1.0 · FULL PRODUCTION BLUEPRINT · ALL RIGHTS RESERVED

SECTION 01

The Grand Vision

This portfolio is not a website. It is a **journey** — a cinematic rite of passage where every pixel serves the narrative. Visitors step through a living, breathing portal that treats the door as a metaphor for knowledge: closed doors represent untapped potential, and each opening reveals mastery. The experience fuses **Renaissance cathedral lighting** with **holographic UI cues**, **ancient mystery** with **ultra-modern digital sophistication**.

"The experience is a rite of passage, not just another scrollable portfolio."

Three Narrative Pillars

- **The Door = Knowledge** — Every gate, hinge, particle and rune supports this metaphor.
- **Ancient Mystery × Ultra-Modern Tech** — Renaissance cathedral lighting meets holographic UI cues.
- **Cinematic Continuity** — The camera never hard-cuts; every section is a single, choreographed take.

Overall User Journey (5 Acts)

ACT 1	ARRIVAL	User lands on page → sees only the Grand Door in a mystical void
ACT 2	SENSING	Door detects presence → veins pulse, particles orbit
ACT 3	OPENING	Door unfolds mechanically → warp-speed particle tunnel
ACT 4	FOYER	User enters circular sanctum → crystalline nav artifacts float
ACT 5	DISCOVERY	Hall of project doors → click to enter case study rooms

SECTION 02

Visual Language & Material Bible

Color Palette (HDR-Ready)

#0B0F2B	Deep Indigo	Primary background — the void
#121212	Charcoal	Secondary background panels
#0E0E0E	Obsidian Gloss	Ground plane reflective surface
#1E9FFF	Electric Blue	Primary emissive — knowledge energy
#E5C16F	Warm Gold	Secondary emissive — wisdom highlight
#00F7FF	Accent Cyan	Bioluminescent vein glow
#E8B4A0	Rose Gold	Luxury door material
#C5C9C6	Brushed Titanium	Material neutral / text

Surface & Material Library

Polished Dark Titanium	Hero door body	PBR roughness 0.15, metalness 0.9, anisotropic highlights
Bioluminescent Filament	Animated vein network	Custom emissive map; speed modulated by user proximity
Black Marble + Gold Veining	Foyer floor inlay	Normal map at 2K, specular reflections
Smart Glass Panels	Data doors & nav runes	Thin-film Fresnel shader, transmission 0.95, IOR 1.52
Obsidian Mirror	Ground plane	Anisotropic, roughness 0.02, metallic 1.0, ray-traced reflections
Ancient Stone	Corridor walls	Micro-pitting + softened edges + gold inlay imperfections

SECTION 03

The Grand Entrance — Hero Door

Initial State (0s – 3s)

- **Camera:** 60mm focal length, dolly at -8 units Z-axis, looking up 15°
- **Environment:** 40,000 floating particles (size 0.01–0.03 units) with Perlin noise drift
- **Door:** 4 units tall × 2.4 units wide, centered. Veins pulse at 0.5 Hz (sin wave)
- **Lighting:** Single spot (12,000 lumens, cone 45°, shadow map 4096px) from 45° high-left
- **Ground:** Reflective obsidian plane mirrors door with slight distortion

Sensing Phase (3s – 4s)

- Trigger: cursor proximity raycast (threshold 5 units)
- Vein pulse accelerates 0.5 Hz → 2 Hz over 500ms (ease-out cubic)
- Emissive intensity ramps ×2.5 — blue responds first, then gold answers
- Sub-bass frequency sweep begins (60 Hz → 30 Hz)

Opening Sequence — The Unfold (4s – 7s)

The door does **not swing** — it **unlocks itself** like a futuristic clockwork mechanism.

DECONSTRUCTION 4.0–5.5s

64 hexagonal plates (Voronoi fracture) extrude 0.3 units with spring physics (damping 0.85, tension 180). Gold filament seams light up. Veins bloom into particle emitters: 500 particles/sec/emitter.

DISSOLUTION 5.5–6.5s

Center 32 plates liquefy into 8,000 metaball particles. Shader: additive blending with motion trails (0.1s persistence). Camera dolly zoom FOV 60 → 120mm while pulling back -2 units Z. God rays appear as light is "revealed by access."

WARP TUNNEL 6.5–7.0s

400 particles/frame form spiral vortex (12 revolutions). Post-processing: chromatic aberration +0.08, barrel distortion +0.15. Screen fades to white at 6.8s. Sound: layered white-noise sweep + reversed choir pad.

SETTLEMENT 7.0s

Camera ease-in deceleration into Foyer position (Z: +12 units). Exposure auto-adjusts to new ambient lighting (HDR bloom threshold 1.2).

Door Asset Specs

- Model: 4 sub-meshes, 80k triangles, rigged with 12 bone pivots for folding panels
- Emissive shader: $\text{mix}(\text{gold}, \text{blue}, \sin(\text{uTime} * \text{speed} + \text{uv.y} * 10.0) * 0.5 + 0.5)$
- Animation curves: overshoot + friction to imply heavy mass
- Audio: descending metallic resonance on fold, crystalline chime on full open

SECTION 04

The Inner Sanctum — Foyer of Ideas

After the warp tunnel, the camera emerges into a warm circular chamber — the **Foyer of Ideas**. No menu bar exists. Navigation is expressed through **floating crystalline artifacts** that embody each section.

Architecture

- 32-sided cylinder room — radius 12m, height 9m
- Ceiling: oculus with animated aurora shader spraying warm light beams
- Floor: polished dark marble with gold radial inlay rings (astrolabe motif)
- Lighting: warm amber key light + cool indigo rim for separation
- Volumetric shafts drift slowly as if the room "breathes"

Navigation Artifacts (No Menu Bar)

ABOUT ME · Irregular geode crystal

Inner glow amber #FFD166, intensity 3. Idle: slow rotation 0.5 RPM, 10-unit hover sin-wave. Hover: scale 1→1.25, glow intensifies, soft chime 432 Hz 0.3s decay.

SKILLS · Torus knot (complexity 5)

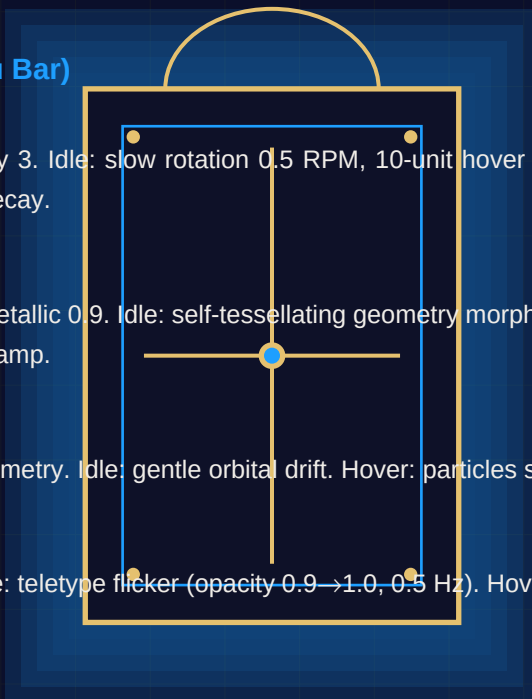
Rose gold PBR, albedo #E8B4A0, metallic 0.9. Idle: self-tessellating geometry morph (Lerp 0.1/s). Hover: morph speed ×5, shader switch to iridescent rainbow ramp.

PROJECTS · Orbiting crystal cluster

Electric blue emissive, instanced geometry. Idle: gentle orbital drift. Hover: particles scatter then reform.

CONTACT · Floating rune ring

Glass material with edge lighting. Idle: teletype flicker (opacity 0.9→1.0, 0.5 Hz). Hover: runes translate into contact glyphs, edge glow cyan.



Interaction Language

- Hover: micro-orbit toward cursor + brightness bloom + subtle pitch-shift sound
- Gaze detection: 0.5s hold triggers navigation (no click required for main nav)
- Click: artifact emits radial wave across floor inlay rings — ripples in stone-light
- Keyboard: same as hover with clean outline pulse (accessibility retained)
- Mouse orbit: ≤10° yaw for parallax depth effect

SECTION 05

Hall of Creations — Project Doors

Scrolling advances the camera along a rail — like a steadicam shot — into the **Hall of Creations**: a grand corridor with arched Gothic ceilings, ancient stone fused with smart-glass inserts. Each project door is a **trailer for its case study** — the interface communicates the project's personality before a click.

Corridor Architecture

- 80 units long, Gothic parametric arched ceiling, walls 2 units apart
- Volumetric fog density 0.15, albedo cyan, with 3 god rays (raymarch steps 128)
- Floor: dark stone with thin luminous seams guiding forward motion
- Each door bay receives a subtle spotlight as user approaches — no arrows needed

Shared Door Behavior Rules

- **Idle**: slow emissive breathing + faint environmental animation
- **Hover**: brighten + unique audio signature + surface detail reveals (normal map intensity rises)
- **Click**: door opens inward → dramatic camera push into project room → `pushState(url/slug)`
- **Return**: small glowing sigil near threshold → smooth close + camera LERP back to corridor

Project Door Design Examples

Each door below represents a project slot. Customize material and color per your actual projects.

DOOR 1 — Fast App / Web App

Material: Brushed chrome + yellow speed lines

Subsonic vibration 0.05 units 22Hz. Hover: neon pulse to max intensity 8, camera rumble. Open: door slides left 0.3s, camera dolly-zoom at 15 units/s. Sound: tight synthetic tick + short whoosh.

DOOR 2 — Luxury / Brand Project

Material: Frosted glass + rose gold filigree

Soft breathing glow sin-wave 0.8Hz. Hover: glass clarifies transmission → 0.98, delicate floral patterns bloom. Open: melts away shader dissolve 0.4s, camera pushes through liquid. Sound: warm chime with long tail.

DOOR 3 — Data / Analytics Project

Material: Smart glass panels 4×4 grid — live mini data viz

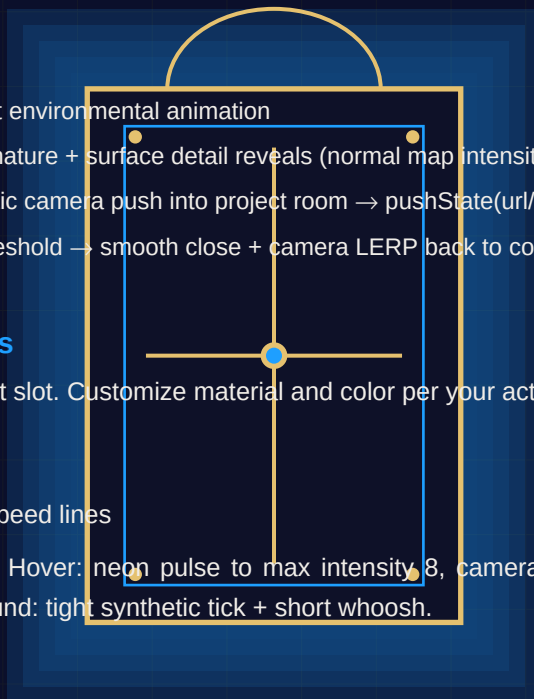
Graphs animate 1000 data points 30fps. Hover: selected panel scales 2×, others dim opacity 0.3. Open: targeted panel fractures into UI elements that become the page. Sound: soft digital crackle + low tonal ping.

DOOR 4 — Creative / Design Project

Material: Vines of glowing code, digital flowers on hover

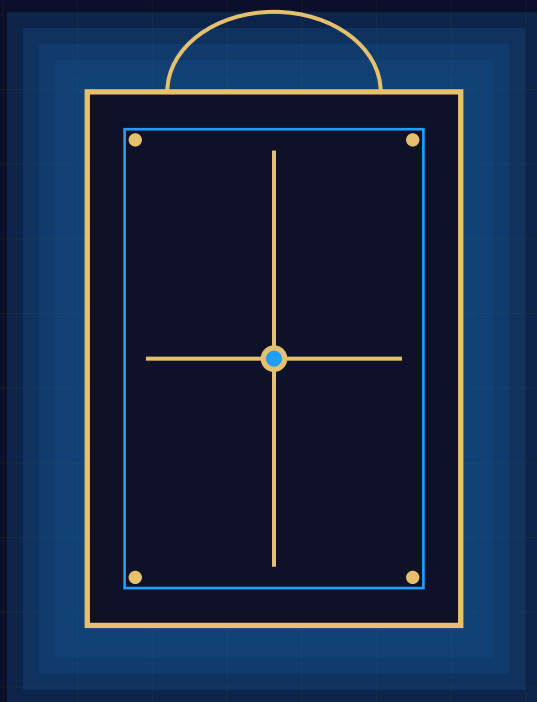
Procedural organic growth animation. Hover: flowers bloom outward in particle burst. Open: vines retract to reveal interior garden. Sound: wind chimes + organic crackle.

DOOR 5 — AI / Tech Project



Material: Swirling neural network patterns in electric purple

Self-organizing node connections animate continuously. Hover: network activates, nodes pulse in cascade. Open: portal effect — circular iris expansion. Sound: digital synth sweep + bass pulse.



SECTION 06

Portfolio Sections Layout

Once the Grand Door opens and the foyer is revealed, the full portfolio structure appears. Here is every section with its content and design approach.

NAV BAR

Floating holographic ribbon at top. Logo left (animated rune), nav links as glowing sigils, no solid background — semi-transparent with blur. Scrolled state: collapses to minimal dot-navigation on the right edge.

HERO — ABOUT ME

Full-viewport section after foyer. Your professional photo in a "portal frame" — circular, with animated electric blue ring and particle halo. Name appears in large serif gold typography with letter-reveal animation. Subtitle and short bio text unfurl like ancient scrolls in a modern HUD. Background: faint rotating geometric tessellation.

SKILLS & LANGUAGES

A constellation of skill icons floating in 3D space. Languages (Python, JS, etc.) appear as glowing orbs of different sizes (size = proficiency). Hover each orb: it expands, rotates, shows proficiency %. Arranged in a galaxy-like cluster with subtle orbital drift animation.

PROJECTS

The Hall of Creations corridor. Each project = a unique door. Click → animated door-open → case study room with: GitHub link button, video demo embed, tech stack pills, challenge/solution/result panels displayed as floating stone slabs or glass panels.

EXPERIENCE / TIMELINE

A vertical "path of light" timeline. Each milestone is a glowing node on the path. Scroll reveals nodes with expanding detail cards that slide out like hidden compartments.

CONTACT

The Contact rune artifact from the foyer links here. Minimal form floats in a circle of particles. Send button triggers a small "message sent" particle burst animation.

Project Case Study Room Structure

- **Hero:** Project name in large emissive typography + hero video/image
- **Overview:** 3 monolith cards — Problem · Solution · Result
- **Tech Stack:** Floating pill badges with animated entry
- **Demo:** Embedded video or live preview in "smart glass" frame
- **Links:** GitHub button (animated door-lock icon), Live Site button
- **Return:** Glowing sigil near the threshold → close door → back to corridor

Technical Stack & Implementation

Core Technologies

Three.js r158	Core WebGL 2 renderer — all 3D scenes, materials, shaders
GSAP 3 + ScrollTrigger	Camera rails, timeline orchestration, scroll-based animations
React / Next.js	Underlying structure, routing, case study pages as separate routes
Custom GLSL Shaders	Emissive veins, Fresnel glass, volumetric fog, particle systems
Howler.js	Spatial audio engine — mastered at -14 LUFS, one bus per ambiance zone
DRACO + KTX2	Compressed geometry & textures — target initial download < 6 MB
Post-processing	UnrealBloom, GodRaysPass, BokehPass (DOF), FilmGrainPass

Post-Processing Pipeline (Stack Order)

1. TAAR — Temporal Anti-Aliasing + Reprojection (16 samples)
2. Bloom — threshold 0.8, intensity 1.5, 5 iterations
3. Volumetric Light — Mie scattering anisotropy 0.8
4. Depth of Field — focal length linked to camera target, bokeh 6 blades
5. Film Grain — 2% intensity, animated UV offset
6. Chromatic Aberration — subtle (<0.02) except during warp transitions
7. Screen-space reflections — obsidian and marble surfaces only

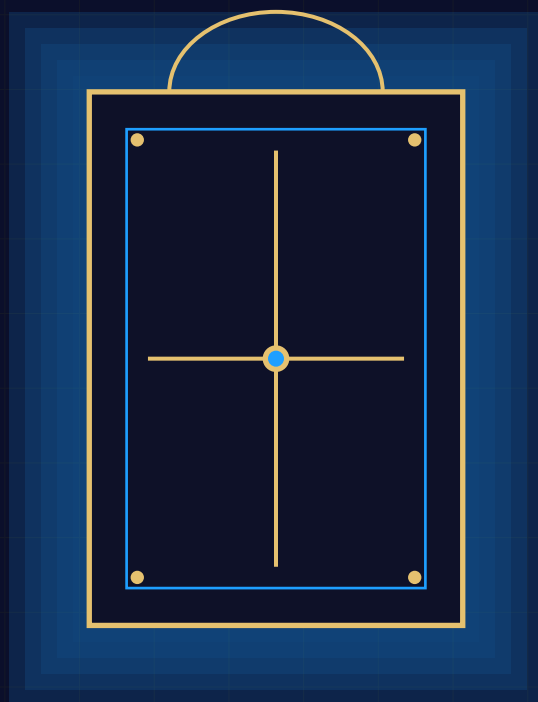
Performance Tiers

HIGH (Desktop / RTX 3060+)	Full post stack + volumetrics + 40k particles + ray-traced reflections
MEDIUM (Laptop / Integrated)	Reduced god rays + simplified reflections + 15k particles
LOW / Mobile	Static background video (H.265 1440p 30fps) + interactive HTML overlay

LOD & Optimization Rules

- LOD switch at 7m, 15m, 30m for all project doors
- GPU instanced particles for warp tunnel and ambient dust

- Baked + dynamic hybrid lighting (baked ambience, dynamic highlights)
- Deferred second scene loading for case study rooms
- Reduced-motion media query: replaces warp with dissolve + camera cut



SECTION 08

Sound Design & Animation Curves

Sound Cue Sheet

Door pulse idle loop
Low-frequency rumble @ 40 Hz + subtle sparkles, loops seamlessly

Sensing activation
Sub-bass sweep 60 Hz → 30 Hz, 500ms duration

Folding segments
Metallic resonance with descending pitch (implies heavy mass)

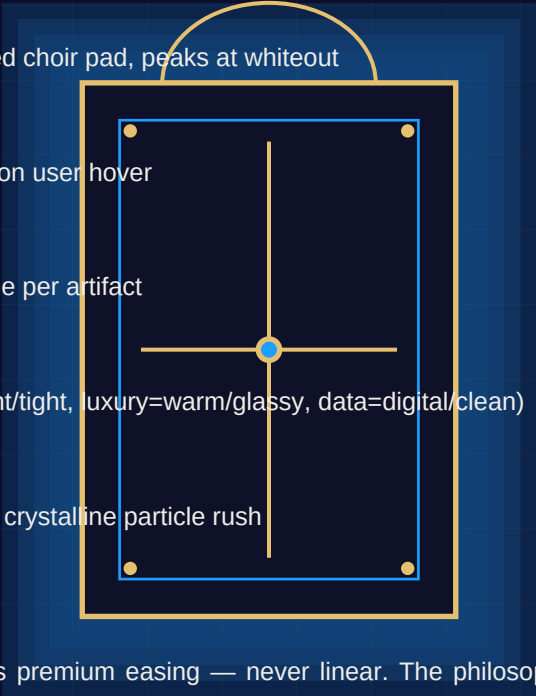
Warp tunnel
Layered white-noise sweep + reversed choir pad, peaks at whiteout

Foyer ambient bed
432 Hz glass harmonics, filter opens on user hover

Crystal artifact hover
Soft chime 432 Hz, 0.3s decay, unique per artifact

Project door hover
Per-door sonic identity (chrome=bright/tight, luxury=warm/glassy, data=digital/clean)

Door open
Mechanism clicks + sub-bass swell + crystalline particle rush



Animation Curve Philosophy

Every animation in the portfolio uses premium easing — never linear. The philosophy: slow anticipation builds tension, decisive action delivers impact, gentle settle implies mass and craftsmanship.

Door veins pulse	<code>sin(uTime * freq + uv.y * 10.0) * 0.5 + 0.5</code>	ease-out cubic ramp
Panel unfold	Spring physics: damping 0.85, tension 180	easeInOutExpo
Warp tunnel	FOV: 40° → 70° over 1.2s	easeInExpo surge
Camera settle	Deceleration into Foyer position	easeOutQuart
Artifact hover	Scale 1 → 1.25 over 0.3s	easeOutBack (overshoot)
Door open click	GSAP scrub to doorOpen timeline	easeInOutCubic

SECTION 09

Accessibility & Production Timeline

Accessibility Essentials

- prefers-reduced-motion: replaces warp with dissolve + fade + camera cut
- ESC key: enables reduced-motion mode at any time during autoplay rail
- Keyboard navigation: Tab through artifacts and doors with clean outline pulse
- Text contrast: adaptive rim lighting behind captions ensures readability
- Screen reader: semantic HTML overlay with aria-labels for all interactive elements
- Mobile: simplified scene + touch gestures replace mouse orbit

8-Week Production Timeline

Week 1	Mood boards, hero door concept sculpt, style frames in Blender Cycles
Week 2	Finalize all shaders & rig; pre-optimize textures (KTX2, DRACO)
Week 3	Build hero scene — Grand Entrance, implement camera rail with GSAP ScrollTrigger
Week 4	Foyer models + rune artifact interaction system; UX tests
Week 5	Corridor generator & all project door variants; case study rooms
Week 6	Post-processing tuning, sound design pass, accessibility mode
Week 7	QA on devices; performance budget target: 60fps on M2 MacBook & RTX 3060
Week 8	Launch, Lighthouse audit pass, A/B test preload variants

Deliverables Checklist

- High-res style frames (Blender Cycles) — hero door, foyer, one project door
- GLB asset pack + texture atlases (2× 4K HDRI env maps)
- Figma UI overlay file (text styles, label components, accessibility contrast ratios)
- Prototype repo with camera rail + main door animation MVP (Three.js)
- Sound stem library (24-bit 48 kHz WAV)
- Performance-tested build: Lighthouse score > 90 on desktop

SECTION 10

AI Prompt Guide — How to Build This

Use the prompts below when working with AI coding assistants (Claude, GPT-4, Cursor, v0) to build each section. Be specific — the more detail you give, the better the output.

Grand Entrance Door (Three.js)

PROMPT TO USE:

Build a Three.js scene with a large dark-metal door centered in a void environment. The door should have animated bioluminescent veins (electric blue + gold emissive shader). On mouse hover, veins pulse faster. On click, animate the door unfolding into 64 hexagonal plates using spring physics (damping 0.85, tension 180), then dissolve into 8000 particles that rush toward the camera in a warp-speed effect with chromatic aberration. Use GSAP for timeline orchestration. Color palette: background #0B0F2B, door #0E0E0E, veins #1E9FFF and #E5C16F.



Foyer Navigation Artifacts

PROMPT TO USE:

Create a Three.js circular foyer room (radius 12m cylinder) with 4 floating 3D artifacts as navigation elements: a geode crystal, a torus knot, a crystal cluster, and a rune ring. Each floats with a gentle sine-wave hover animation. On mouse hover, each artifact scales up 1→1.25, increases emissive intensity, and plays a unique sound via Howler.js. Clicking triggers a camera LERP transition to the target section.



Project Doors Corridor

PROMPT TO USE:

Create a scrollable Three.js corridor 80 units long with Gothic arched ceiling. Line the walls with 5 project doors, each with unique materials: (1) brushed chrome with yellow speed lines, (2) frosted glass + rose gold filigree, (3) transparent smart glass with animated data graphs, (4) glowing vine code patterns, (5) swirling neural network purple portal. Scroll moves the camera forward along a rail using GSAP ScrollTrigger. Each door brightens on hover and opens with a camera push-in on click.



Full Portfolio Structure (React + Three.js)

PROMPT TO USE:

Build a single-page React portfolio with Next.js and Three.js (react-three-fiber). Sections: (1) Hero door scene with opening animation, (2) About Me with parallax photo and animated name reveal, (3) Skills as floating 3D orb constellation, (4) Projects as doors corridor, (5) Timeline as vertical path of light, (6) Contact with particle burst on submit. Global color palette: #0B0F2B background, #E5C16F gold accents, #1E9FFF blue emissive. All animations use GSAP with premium easing. Include reduced-motion fallback.

THE FINAL FEELING

Visitors who arrive at this portfolio should feel they are entering a secret vault of digital artifacts — guided by light, sound, and mechanism rather than UI chrome. They experience your work as rooms of knowledge behind doors, each with its own atmosphere and promise.

This is not a website. It is a rite of passage. Every element — from the unfolding of the grand door to the individual project portals — tells a story of imagination, craftsmanship, and innovation. Visitors will leave not only with an understanding of your work, but with a memorable experience that sets you apart as a creator at the intersection of art and technology.

LET'S OPEN THE DOOR.

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